

SHEFFIELD, Australia

WMO: 949550

Lat: 41.3886S

Lon: 146.3219E

Elev: 295

StdP: 97.84

Time Zone: 10.00 (AUS)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.7	1.6	-3.0	3.0	5.1	-2.0	3.3	5.1	12.6	8.0	11.2	8.0	1.6	340	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.2	25.3	17.5	23.4	16.7	21.9	16.0	19.0	23.1	18.0	21.3	17.1	20.5	4.5	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.6	13.0	20.2	16.7	12.3	19.6	15.7	11.6	18.7	55.1	23.4	51.9	21.3	49.3	20.4	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.7	9.3	8.2	-1.0	29.3	0.6	2.7	-1.5	31.2	-1.8	32.8	-2.1	34.3	-2.6	36.2		
(5)			-1.6	21.0	0.8	0.9	-2.1	21.6	-2.6	22.1	-3.0	22.6	-3.6	23.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.2	16.0	15.9	14.3	11.6	9.4	7.8	7.2	7.4	8.5	10.1	12.4	13.9		
	DBStd	3.89	2.83	2.62	2.73	2.36	2.09	1.85	1.67	1.87	1.99	2.31	2.56	2.60		
	HDD10.0	374	0	0	1	11	37	68	88	81	53	27	6	2		
	HDD18.3	2632	83	75	127	202	278	314	346	338	294	256	178	141		
	CDD10.0	811	187	166	136	59	18	3	1	2	8	30	78	122		
	CDD18.3	25	11	8	4	0	0	0	0	0	0	0	0	2		
	CDH23.3	183	100	39	19	1	0	0	0	0	0	0	5	18		
	CDH26.7	32	25	3	2	0	0	0	0	0	0	0	0	2		
(14) Wind	WSAvg	3.5	3.8	3.4	3.2	3.0	3.2	3.0	3.3	3.6	3.7	3.7	3.7	3.9		
(15) Precipitation	PrecAvg	1003	62	51	48	65	88	110	131	130	107	75	74	61		
	PrecMax	1573	190	122	117	186	225	296	221	285	196	188	172	153		
	PrecMin	505	3	6	3	19	9	12	55	32	27	13	12	5		
	PrecStd	210	52	28	33	38	55	57	46	62	46	41	42	41		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.6	26.7	25.6	20.7	16.6	14.2	13.0	14.2	16.7	20.4	23.7	25.8		
		MCWB	17.7	18.3	18.4	15.0	12.6	11.0	10.0	10.6	12.0	14.2	16.0	17.5		
	2%	DB	25.7	24.4	22.9	18.5	15.0	12.8	11.8	12.8	14.8	17.9	21.3	22.7		
		MCWB	17.4	17.7	16.6	14.0	11.9	10.4	9.3	9.9	10.8	13.3	15.3	15.9		
	5%	DB	23.6	22.7	20.8	17.1	14.0	12.0	11.0	11.8	13.5	16.1	19.2	21.0		
		MCWB	16.6	16.5	15.8	13.4	11.5	10.0	9.1	9.5	10.1	12.1	14.2	15.1		
	10%	DB	21.7	21.2	19.2	15.8	13.1	11.2	10.3	10.9	12.4	14.8	17.6	19.3		
		MCWB	15.8	15.7	14.7	12.6	11.0	9.4	8.6	9.0	9.6	11.4	13.3	14.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	20.7	20.2	19.3	16.5	14.1	12.2	11.3	12.0	13.1	15.6	17.8	18.7		
		MCDB	26.8	24.4	24.0	18.7	14.8	12.9	11.9	12.9	14.9	18.0	21.3	23.6		
	2%	WB	19.0	18.8	18.0	15.2	13.0	11.2	10.3	10.9	11.9	14.2	16.2	17.3		
		MCDB	23.5	22.3	20.9	17.1	14.1	12.1	11.0	11.9	13.6	16.5	19.6	20.9		
	5%	WB	17.8	17.7	16.9	14.2	12.2	10.5	9.6	10.0	10.9	13.1	15.0	16.3		
		MCDB	21.5	20.9	19.7	16.1	13.3	11.5	10.5	11.2	12.8	15.3	18.1	19.6		
	10%	WB	16.7	16.8	15.8	13.3	11.4	9.7	9.0	9.3	10.0	12.0	13.9	15.1		
		MCDB	20.5	20.0	18.2	15.2	12.7	10.8	10.0	10.5	11.9	14.3	17.0	18.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.2	9.5	9.0	8.1	6.9	6.3	6.1	6.6	7.4	8.4	9.1	9.5		
		MCDBR	12.8	11.2	10.6	9.5	7.5	6.7	6.4	7.4	8.6	10.0	11.3	11.9		
		MCWBR	6.4	5.7	5.7	5.8	5.4	5.0	4.9	5.4	5.8	6.7	6.8	6.6		
	5% WB	MCDBR	11.3	9.8	9.6	8.2	6.3	5.7	5.5	6.5	7.8	9.0	9.8	10.1		
		MCWBR	6.8	5.8	6.2	5.7	5.2	4.7	4.7	5.4	5.9	6.7	6.5	6.6		
(40) Clear-Sky Solar Irradiance	taub		0.333	0.328	0.317	0.307	0.293	0.287	0.287	0.301	0.322	0.323	0.326	0.325		
	taud		2.528	2.557	2.582	2.591	2.604	2.605	2.614	2.573	2.499	2.512	2.531	2.546		
	Ebn at Noon		995	974	942	884	830	802	830	877	919	969	996	1009		
	Edh at Noon		108	100	89	76	63	58	62	76	95	104	107	108		
(44) All-Sky Solar Radiation	RadAvg		6.92	5.90	4.46	3.02	1.93	1.50	1.63	2.42	3.56	5.14	6.20	6.76		
	RadStd		0.45	0.25	0.19	0.20	0.15	0.15	0.09	0.20	0.26	0.30	0.46	0.47		

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)		+0.31	N/A	N/A	+0.68	+0.35	+0.43	N/A	-98	+106	+12

Nomenclature: See separate page